data quality and availability tests of public seismometer data in europe

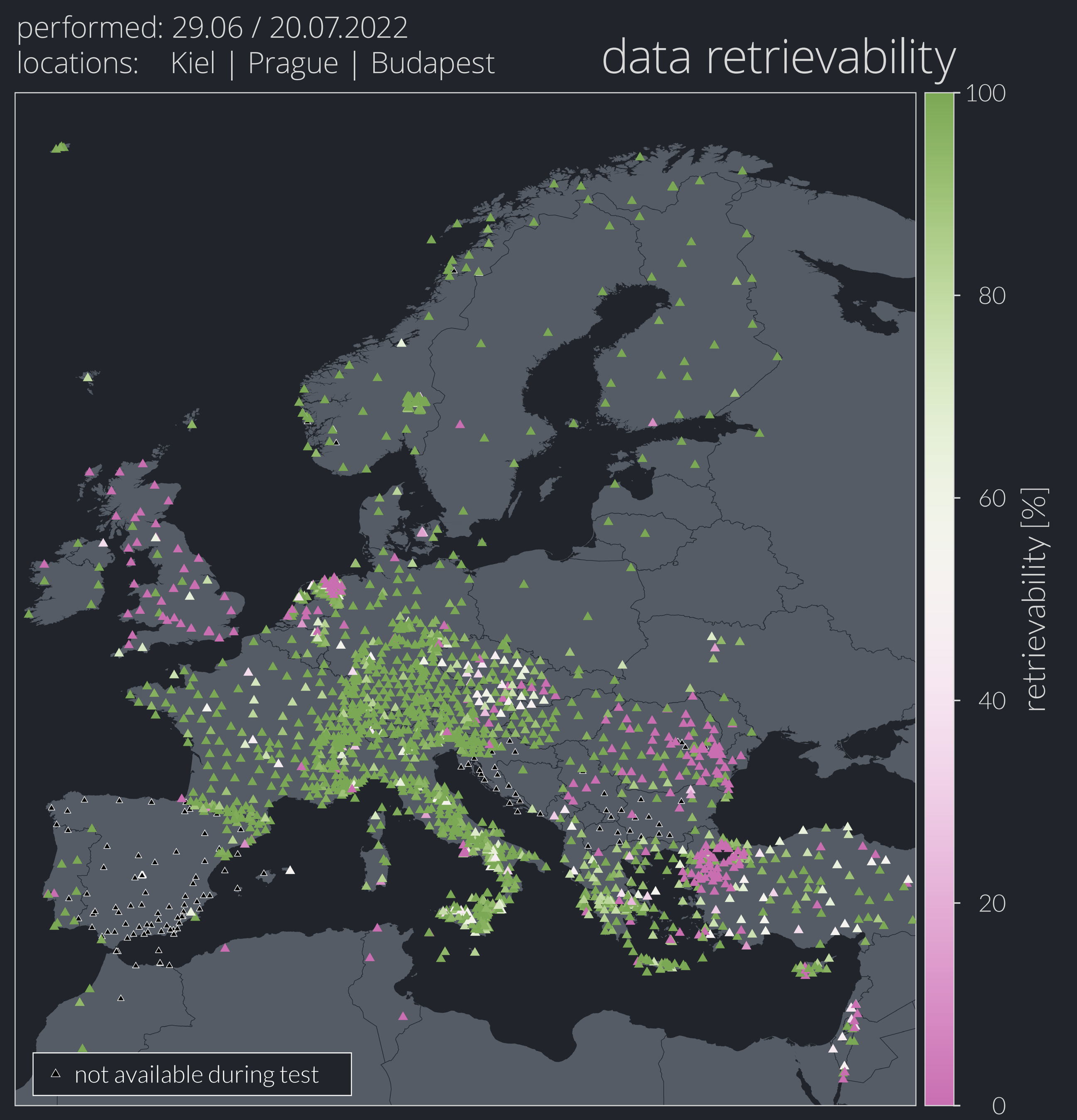
F. Eckel[1] | J. Stampa[1] | M. Timkó[2] | L. Vecsey[3] | P. Kolínský[3]

[1]Institut für Geowissenschaften, Christian-Albrechts-Universität, Kiel, Germany
[2]EPSS Kövesligethy Radó Seismological Observatory, Budapest, Hungary
[3]Institute of Geophysics, Czech Academy of Sciences, Prague, Czech Republic

felix.eckel@ifg.uni-kiel.de
@FelixEckel

Many European countries agreed to share their seismic waveform data via the European Integrated Data Archive infrastructure (**EIDA**) where data from **thousands of seismic stations** is available. With these **large data sets**, manual data quality checks become more and more unfeasible.

Due to **rising traffic** at the data centers, errors are more likely to occur. Identifying stations and networks that are **more likely to be not available** upon requests or provide **erroneous data** is important in terms of quality control.



Data retrievability tests are performed by requesting **random** hourly data chunks within one year for all stations at the data centers (**EIDA node**). The request is forwarded by a routing client.

For every station in EIDA the amount of data that can be **successfully downloaded** is protocolled.

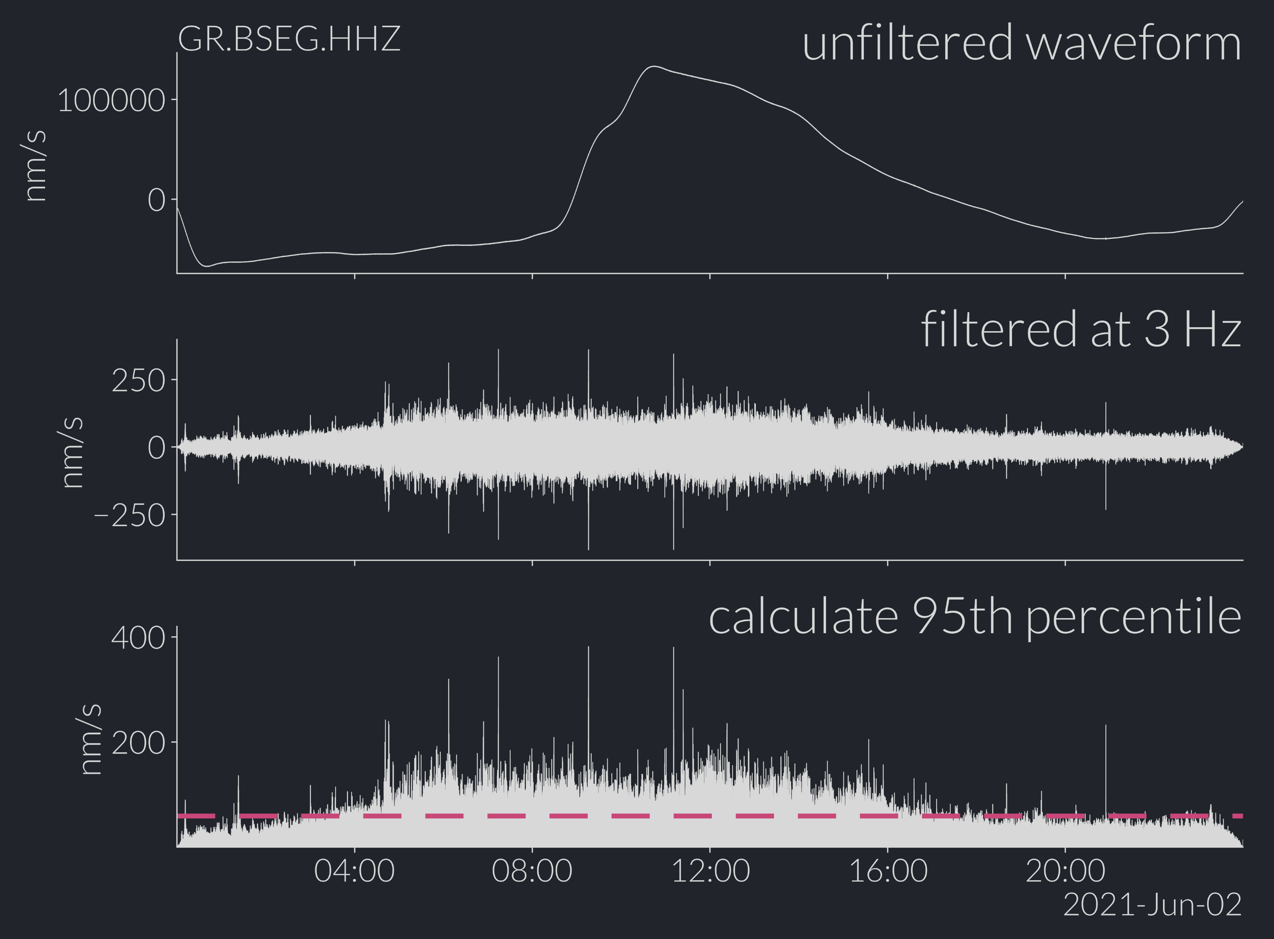
The test is simultaneously performed from **three different locations** (Kiel, Prague, Budapest). For each station the **retrievability** is calculated as the ratio of the requested data from all indivdiual tests and the downloaded amount of data.

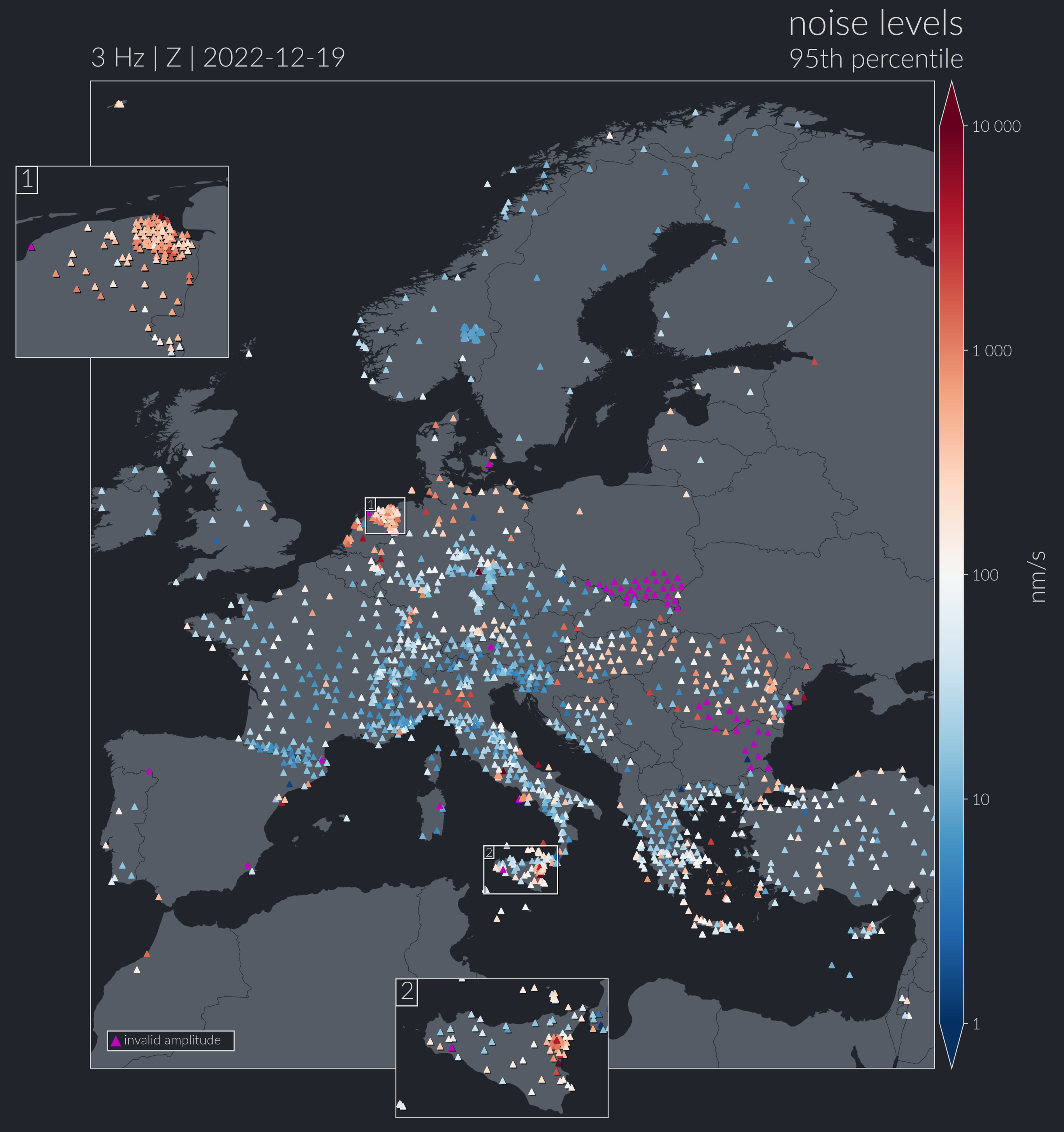
Stations that are known to EIDA but **cannot be downloaded** appear in **magenta** colors. Stations for which every requested data chunk could be **successfully downloaded** appear in **green**.

Data quality checks are vital to avoid processing of false data. Criteria for data quality depend on the methods being used. Here, a **noise level analysis** is used to focus on reported amplitudes.

The average noise level at each station is calculated as the **95th percentile** of the **filtered absolute amplitudes**.

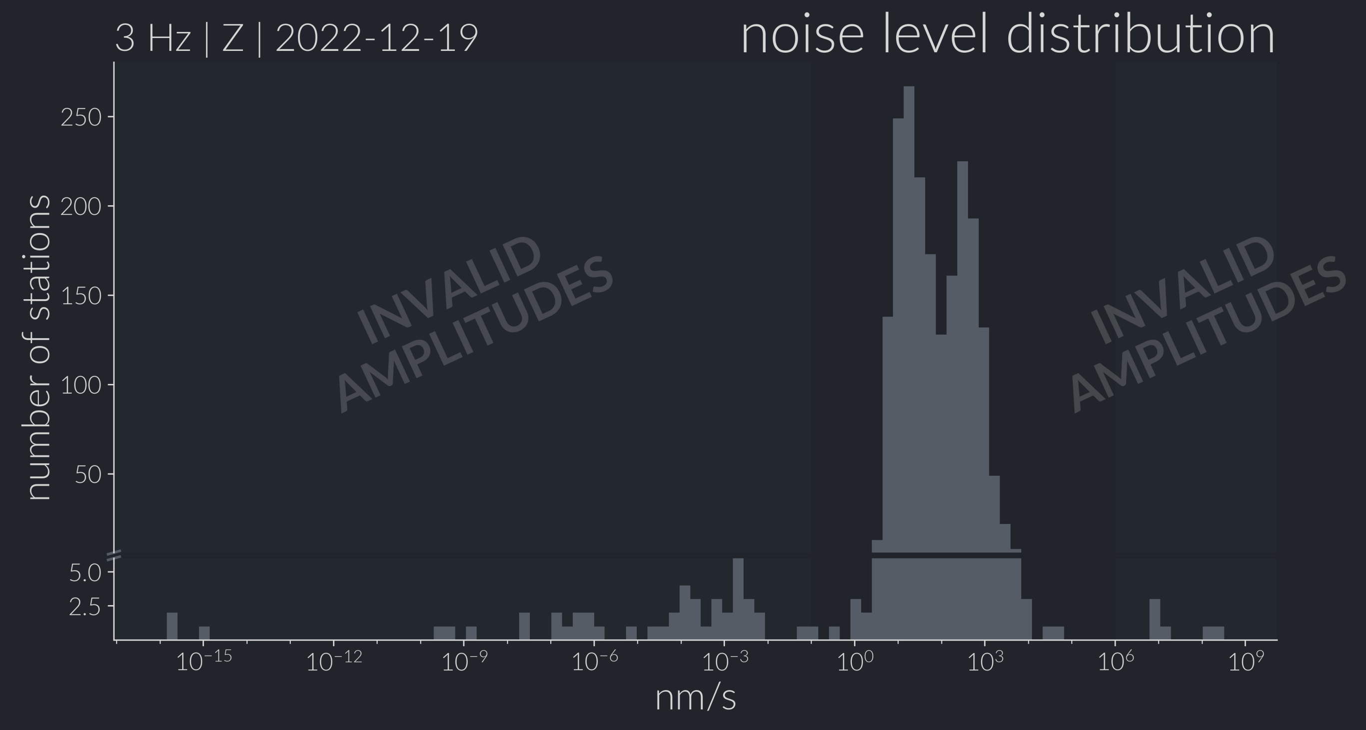
Comparing both the absolute amplitude values and the noise levels at neighboring stations, **false sensitivity values** or other **metadata problems** can be identified.

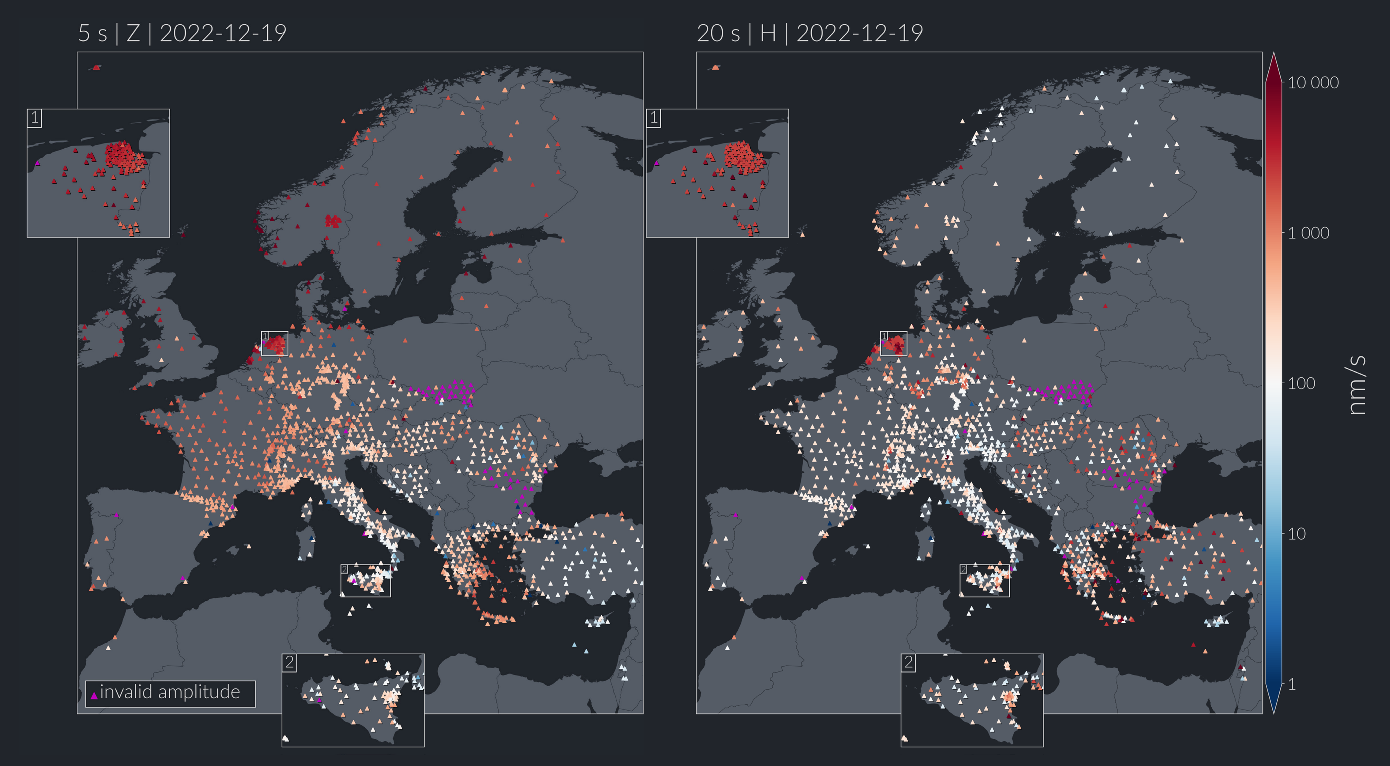




Average noise levels indicate structural variations over Europe, e.g. sedimentary basins (Po basin, molasse basin) or orogens (Alps, Apennines).

Stations with problems are identifiable by **outliering amplitudes**. These areas are marked as "invalid" in the histogram and colored in magenta on the map. Reasons for these outliers might be false metadata reportings or site effects.





Filtering in **frequency bands** with decreasing central frequency can reveal stations that **perform poorly** at long periods. Additionally the effect of the **microseism** band on the data quality becomes intuitively visible. Watch this **video** to see the effect:

Abstract

Sharing is encouraged

Christian-Albrechts-Universität zu Kiel
Mathematisch-Naturwissenschaftliche Fakultät